

APPENDIX: STUDY DOCUMENTATION AND REPRODUCIBILITY PACK

This appendix provides the formal documentation, data structures, and procedural frameworks for the evaluation of the Output-Driven Multilingual Learning (ODML) framework. All materials are prepared in accordance with the reporting standards of Q1 Second Language Acquisition (SLA) and psycholinguistic journals (e.g., Studies in Second Language Acquisition, Language Learning, Brain and Language).

SECTION 1: DATA DICTIONARY & CODEBOOK

This section defines the structure, variable types, and measurement scales for the consolidated results dataset (`results_data.csv`).

1.1 Dataset Overview

- Filename: `results_data.csv`
- Format: Comma-Separated Values (.csv)
- Unit of Analysis: Group-level aggregate summary (Mean and Standard Deviation) per experimental condition, structured by target manuscript table and linguistic/cognitive measure.
- Total Rows: 8
- Total Columns: 9

1.2 Variable Definitions

Variable Name	Storage Type	Measurement Level	Allowed Values / Range	Description
Table	Character	Nominal	Table_2 , Table_4 , Table_5	Identifies the target table in the manuscript where

Variable Name	Storage Type	Measurement Level	Allowed Values / Range	Description
				these descriptive statistics are reported.
Measure	Character	Nominal	Linguistic_Accuracy , Hesitation_Markers , Self_Repair , NASA_TLX	The dependent variable or cognitive/ linguistic metric being evaluated.
Group	Character	Nominal	Experimental , Control	The intervention condition. Experimental denotes the ODML group; Control denotes the Input-First group.
Pre-test_M	Numeric	Continuous	≥ 0	Sample mean at the baseline/pre-test assessment.
Pre-test_SD	Numeric	Continuous	≥ 0	Standard deviation of the sample at baseline/pre-test.
Immediate_Post_M	Numeric	Continuous	≥ 0 or NA	Sample mean at the immediate post-test/post-intervention phase.
Immediate_Post_SD	Numeric	Continuous	≥ 0 or NA	Standard deviation of the sample at the immediate post-test phase.
Delayed_Post_M	Numeric	Continuous	≥ 0 or NA	

Variable Name	Storage Type	Measurement Level	Allowed Values / Range	Description
Delayed_Post_SD	Numeric	Continuous	≥ 0 or NA	Sample mean at the delayed post-test phase (long-term retention). Standard deviation of the sample at the delayed post-test phase.

1.3 Measure-Specific Details

1. Linguistic Accuracy (**Linguistic_Accuracy**)

- Target Table: Table_2
- Metric Unit: Mean rating scale (0 to 5, where 5 indicates native-like target language grammatical and lexical accuracy).
- Temporal Coverage: Measured at Pre-test, Immediate Post-test, and Delayed Post-test.

2. Hesitation Markers (**Hesitation_Markers**)

- Target Table: Table_4
- Metric Unit: Rate per 100 words (frequency count normalized by total produced words).
- Temporal Coverage: Baseline intervention task phase (Pre-test column values). Post-test measures are excluded as this represents an online processing difficulty metric during initial production.

3. Self-Repair Attempts (**Self_Repair**)

- Target Table: Table_4

- Metric Unit: Rate per 100 words (frequency count of self-initiated corrections).
- Temporal Coverage: Baseline intervention task phase (Pre-test column values).

4. Cognitive Load (**NASA_TLX**)

- Target Table: **Table_5**
- Metric Unit: Index score (0 to 100, where higher scores represent higher perceived cognitive workload).
- Temporal Coverage: Baseline/Intervention Phase (reported in **Pre-test** columns) and Immediate Post-test. Delayed post-test is not applicable (**NA**).

SECTION 2: STUDY PREREGISTRATION FRAMEWORK

2.1 Research Questions

1. Does the sequencing of output before input (ODML: Output → Gap Detection → Input → Production) result in significantly higher linguistic accuracy and long-term retention compared to an Input-First curriculum?
2. Does the ODML framework facilitate oral fluency by reducing speech onset latency over time?
3. Does the ODML framework induce a "desirable difficulty" effect, characterized by elevated cognitive load and higher rates of hesitation/self-repair during the learning phase, followed by reduced cognitive load and superior performance at delayed post-tests?

2.2 Hypotheses

- Hypothesis 1 (Linguistic Accuracy): There will be a significant interaction between Group and Time on linguistic accuracy. The Experimental group

will exhibit steeper gains from pre-test to immediate post-test and show significantly higher retention scores at delayed post-test.

$$\text{Accuracy Gain}_{\text{Experimental}} > \text{Accuracy Gain}_{\text{Control}}$$

- Hypothesis 2 (Speech Onset Latency): The Experimental group will show a more pronounced decrease in speech onset latency from baseline to delayed post-tests.
- Hypothesis 3 (Cognitive Load): During the initial intervention, the Experimental group will report higher perceived cognitive load (NASA-TLX) than the Control group, but lower load at immediate post-test.
- Hypothesis 4 (Online Processing): The Experimental group will display a higher frequency of hesitation markers and self-repair during the baseline intervention task.

2.3 Experimental Design & Methods

- Design: Randomized, parallel-group, pretest-posttest-delayed posttest design.
- Blinding: Double-blind (participants and external evaluators).
- Target Sample Size: $N = 60$ (CEFR B2 baseline).
- Duration: 4 weeks (8 sessions).

2.4 Statistical Analysis Plan (SAP)

Outcomes will be analyzed using Linear Mixed-Effects Models (LMMs). * Fixed Effects: Group, Time, and Group $\times$ Time interaction. * Random Effects: By-participant random intercepts. * Significance Threshold: $\alpha = 0.05$.

SECTION 3: DATA MANAGEMENT PLAN (DMP)

3.1 Standards

- File Formats: Raw audio in WAV; tabular data in CSV/TSV.
- Security: Data hosted on password-protected institutional servers. No PII stored; participants identified by alphanumeric ID.

3.2 Quality Control

- Validation: Automated range checks for all quantitative outputs.
- Acoustic: Latency values < 100ms flagged for manual inspection.
- Outliers: Handled via model-based filtration or log-transformation.

3.3 Version Control

- Protocol: [v1.0_RAW] → [v1.1_CLEANED] → [v1.2_ANALYSED] .
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SECTION 4: REPRODUCIBILITY GUIDE

4.1 Required Inputs

1. LMM_Results_Summary.csv : Contains model estimates (β , SE, t , p).
2. results_data.csv : Contains descriptive group means/SDs.

4.2 Reproduction Steps

- Table Generation: Filter LMM_Results_Summary.csv by dependent variable. Format estimates to APA 7th edition (2 decimal places for coefficients/CIs, 3 for p -values).
 - Descriptive Summaries: Pivot results_data.csv to format means and standard deviations as unified cells (e.g., 3.12 ± 0.42).
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SECTION 5: FIGURE GENERATION

The primary interaction effect on linguistic accuracy is visualized by plotting mean performance across time points, using error bars to represent the standard deviation for each condition. Statistical trend lines reflect the differential retention trajectories between the ODML and Input-First groups.